

How the principle works in this build:

What already happens: at every level start the game does exactly "stop the CD audio, read data, start the music again":

- command 54 stops the track (a data read stops it anyway on the Sega CD);
- commands 38 and 46 load the level pack into the cart, with the loading screen;
- command 45 starts the new level's song.

The idea adds loading **ahead**: the next level, or a cutscene, goes into the cart during a pause the player wouldn't notice anyway (the level-end jingle, the score screen, the planet screen). It then sits there ready, and the next level starts without a loading screen.

The room question:

Cart area	Size	Holds
0x000000–0x06E000	0.43 MB	the engine
0x06E000–0x100000	0.57 MB	free
0x100000–0x280000	1.5 MB	common pack (menus, fonts, panel, all planet screens)
0x280000–0x3F0000	1.4 MB	the current level's pack (0.5–1.0 MB)
0x3F0000–0x400000	64 KB	settings and saves area

A second level slot for the preloaded level needs up to about 1 MB, and only 0.57 MB is free. Ways to make room:

1. **Slim the common pack.** About half of it is menu and planet graphics that aren't needed during a level, and only one planet screen is needed at a time.
2. **Smaller level packs.** Pictures that repeat between the levels of the same planet could be shared: each planet's two levels use the same tile set and sprites, but each pack carries its own copy.
3. **Preload only within a planet.** Level 2 of a planet needs only its own small level file once level 1's pack is in place; the tiles and sprites are already there. That alone removes the loading screen for every second level, with no extra room at all.

How much time it saves (an estimate, from the loading speed in the earlier videos):

- The data pack (38 chunks of 32 KB, about 1.2 MB) took about 40 seconds. That's roughly **1 second per 32 KB chunk**, about 30 KB/s.
- **Before:** every level loaded its own pack of about 0.9 MB, so ~28 s per level. Across all 79 levels that's about **37 minutes of loading**.
- **Now:** one pack per planet, about 1.0–1.5 MB, so ~30–45 s, but only once for both levels. 39 loads come to about **22 minutes**.
- **Saved:** roughly **15 minutes over the whole game**, about 20 seconds per planet. The second level of each planet starts at once.

Can it be improved? Yes, and the biggest lever isn't the pack size but the loading speed. The Sega CD reads about 150 KB/s, and we get about 30. Most of the time is lost between chunks, not in reading. Ideas, best value first:

1. **Bigger chunks** (Sub-CPU change).
 - Each 32 KB chunk is a separate command. Between commands the drive likely stops, then has to find its place on the disc again; a seek can take a few hundred milliseconds.
 - Reading 64 or 128 KB per command (more sectors at once into Word RAM, then several 68000 copies) cuts that overhead by 2–4 times.
 - It's an edit to the proven command 46, so it needs careful testing, but it could roughly **halve or quarter every load**: boot, planets, everything.
2. **Leave out data the console never reads** (simple, PC side only):
 - **Tile set raw data:** each planet pack still carries the original BLOCKS file, about 140 KB, although the ready-made tile set is used. The game only reads its colour palettes from that file. Keeping only the palette part saves ~140 KB per planet.
 - **Player sprites:** every planet pack carries its own copy of MAINCHAR.000 (156 KB), which is the same for all planets. Storing it once in the common pack saves ~156 KB per load (and 6 MB of disc), if the common pack has room for it.
 - Together, roughly **25–30% less** to load per planet.
3. **Compression** (Sub-CPU change, more work):
 - Tile and sprite data typically pack to a half or a third of their size.
 - The Sub-CPU would unpack each chunk before the 68000 copies it into the cart, because the SH2 can't write the cart.
 - Combined with 1, this would make loading several times faster.
4. **Preloading in the background** (the second cart slot): loads the next planet during level-end and score screens, so the player sees no loading at all for most transitions. Points 2 and 3 make the packs small enough to fit a second slot.

The red screens after dying or changing levels (out of memory, 46,272 bytes).

- **Cause:** at every level start, the game decoded the status panel graphics (PANEL.000) into a 46KB block of memory. After a few level starts, the 32X's 218KB of memory is too fragmented to find 46 KB in one piece.
- **Not a leak:** on the PC with the 32X code paths, loading a level a second time leaves only 240bytes more in use.
- **Fix:** the panel now comes ready-made from the cart, like the tile sets, so that 46 KB block is gone. The new save method (below) also frees 8 KB of memory.
- The second photo (156,736 bytes) is from the older build that had the Makefile problem; that one was fixed in J8.

Saves: they now go to the RAM cart's save area, like Tyrian and OpenLara (one 4 KB piece per file: settings and the 4 save slots). Two reasons the old method failed:

- the Sega CD's internal backup RAM is shared with every other game, and a Jazz save is up to 2KB;
- the code cut the settings file's name, OPENJAZZ.CFG, to 11 characters, so the settings were never found again. That was the bug.

Test writing and reading back the settings and a 2KB save here.

The other points:

1. **White bars** above and below the picture: those 24 border lines showed palette colour 0, which is white in some levels. They now show the darkest colour of the current palette.
2. **Music and sound volume:** CD audio and the Sega CD's sound chip have no volume control on our path. The set-up sliders now work as on/off: 0 = off, anything above = on.
3. **Sound effects** are much quieter (about 40% of before) and stay below full scale, which should remove the distortion.
4. **Keyboard:** gone from the set-up menu, together with joystick and video.
5. **Buttons:** the set-up menu now has one entry, "buttons: a jump b fire", and selecting it switches to "a fire b jump". C always jumps.

Frame rate: it looks unchanged. **The second SH2** drawing half the picture didn't help noticeably, so drawing isn't where most of the time goes. Rather than guess again, the next step is a small timing display on the console showing how long each part of a frame takes. The real cure for scrolling is still the separate scrolling project.

The questions:

1. **Bugs:** the photos showed one pattern: memory running out after several level starts. I removed the biggest cause. Others probably remain; the red screen names them.
2. **Skipping the load when the data is already in the cart: yes, for the next round.**
 - At start-up the SH2 checks the cart against the disc's identity mark and a checksum over the cart content, which takes a fraction of a second.
 - If both match, the 40-second engine copy and the common pack load are skipped.
3. **Holding a button combination at power-on** will force the full load, for when the cart's content has been disturbed. Wiping the cart is then never needed.

Add J12's planet packs

```
cd /mnt/s/KoboPort/sega-toolchain-12.1/sega
unzip -o -q OpenJazzCD32X/OpenJazzCD32X_J13.zip
cd OpenJazzCD32X
grep -c "loadRLEPacked" oj/src/jj1/level/jj1levelload.cpp
grep -c "SAVE_AREA" port/mars_syscalls.c
./build.sh 2>&1 | tee build_log.txt
tail -16 build_log.txt
```

The grep lines print **1** and **2**; I checked them on a fresh copy.

On the console:

- several levels and deaths in a row;
- save, switch off, switch on, load;
- the sound effects' volume;
- the set-up menu's button swap.